

RESILIENCE & NETWORK SCIENCE

Operations Research + Resilience Layer

Dashboard 4 — Fragility • Network Criticality • Resilience Economics • Research Layer

RANKED NODES

45

operational network

CRITICAL/OVERLOADED

10

risk regimes

AVG FRAGILITY

0.228

node-level metric

AVG CRITICALITY

0.749

network metric

TOP RESILIENCE/\$

0.0391

investment signal

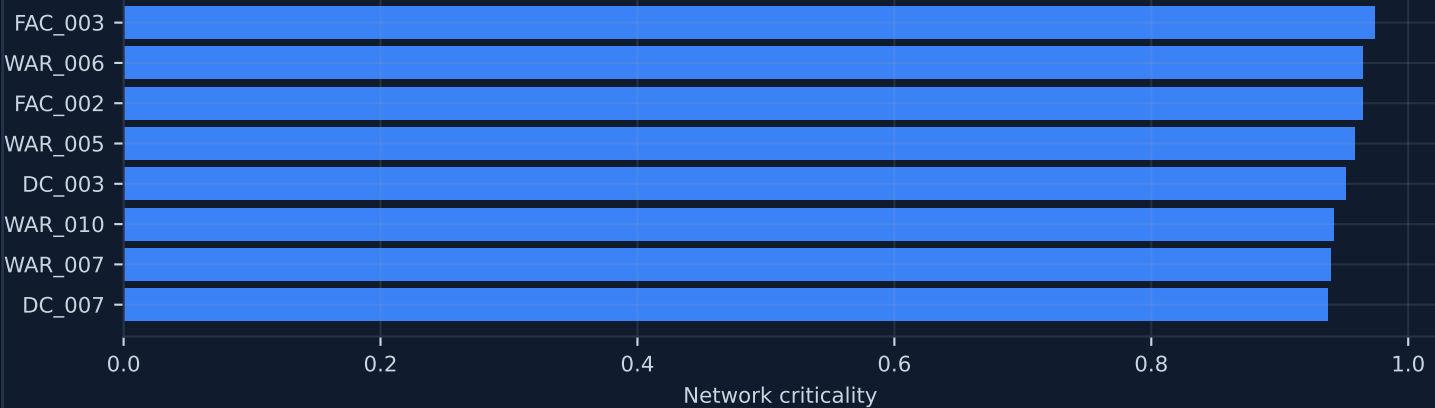
TOP NODE

WAR_012

Warehouse

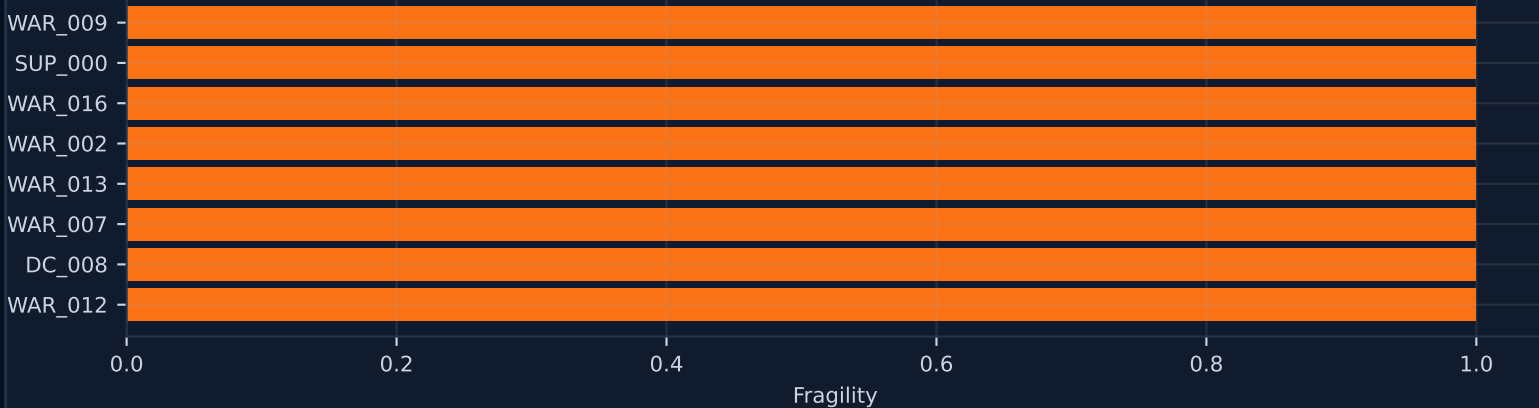
Network Criticality by Node

Structurally important nodes in the supply chain.



Near-Critical Fragility by Node

Operational fragility derived from stability margins.



Resilience Economics

Where investment has the strongest expected return. Bubble size = resilience per dollar



Research Translation Advanced layer demonstrated

From near-critical theory to industrial decision support.

- ✓ Network Science: criticality and structural importance
- ✓ Resilience Engineering: fragility and collapse-risk logic
- ✓ Operations Research: intervention prioritization
- ✓ Decision Systems: risk translated into investment ranking
- ✓ Research Layer: near-critical systems applied to Digital Twins

Core message

The platform does not only report KPIs; it identifies systemic risk.